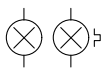
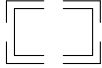
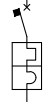
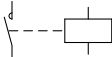
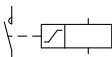
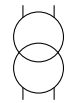
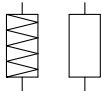


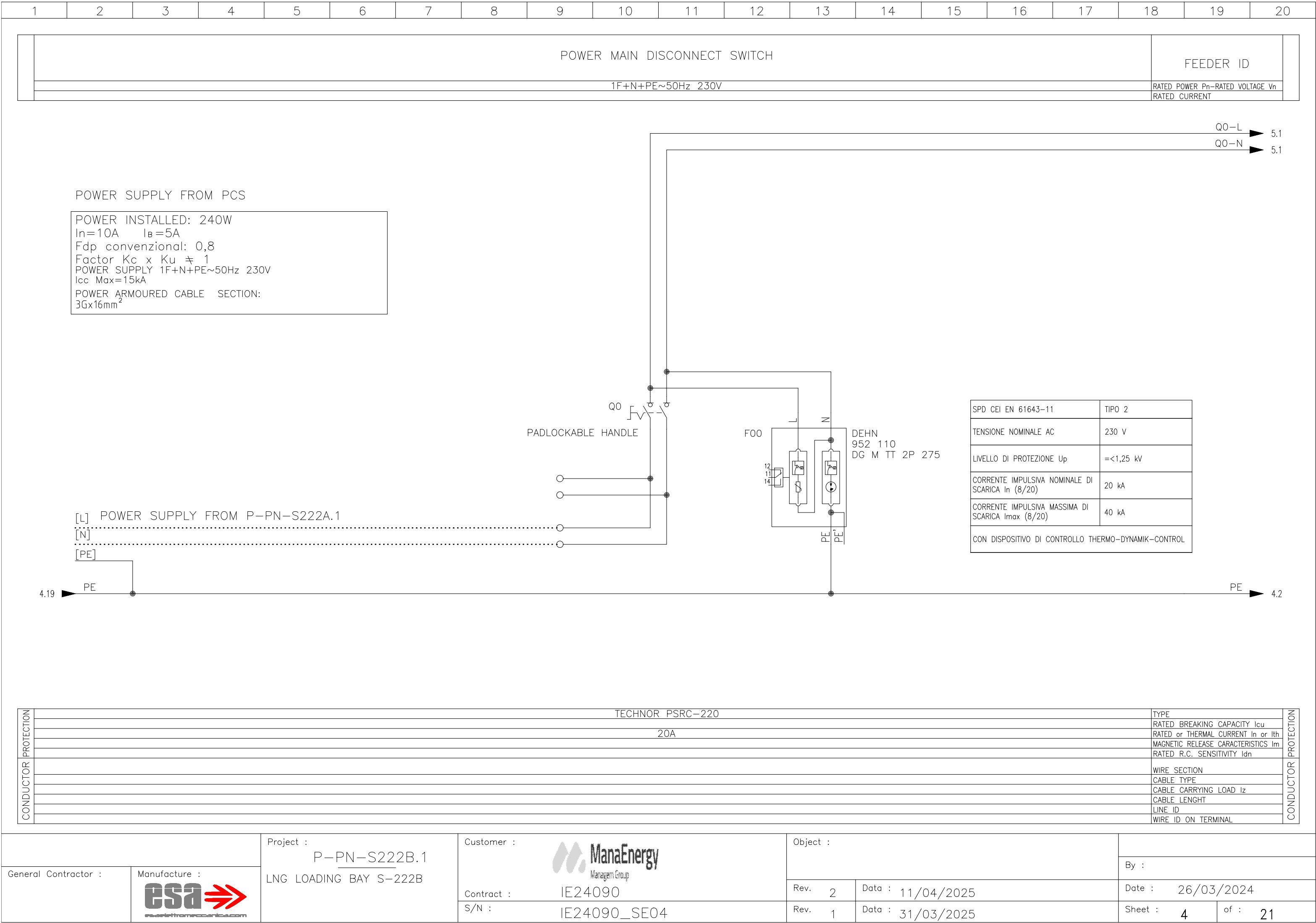
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																				
SAFETY CRITICAL ELEMENT LIST																				
IFI	2	31/10/2025	AS-BUILT				ADF	GDP	IC											
IFI	1	29/01/2025	GENEERAL REVIEW				ADF	GDP	IC											
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC											
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.											
	Rev. Index	Date	Description				Prepared	Checked	Approved											
Client							Project Title													
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT													
Contractor							Provision of a m-LNG plant for production for ManaEnergy													
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																				
Engineering																				
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																				
Facility Name			Location				Scale		Sheet of Sheets											
TENDRARA			MOROCCO				A4		1 of 21											
Document Title					Doc. Code															
P-PN-S222B.1					ELD01-ING-FR01															
LNG LOADING BAY S-222B					Vendor Doc. code IE24090_SE04_P-PN-S22B.1															
			Project : P–PN–S222B.1				Customer : ManaEnergy Managem Group				Object :									
General Contractor :		Manufacture : esa esaelettromeccanica.com		LNG LOADING BAY S–222B				Contract : IE24090				Rev. 2		Data : 11/04/2025				Date : 26/03/2024		
								S/N : IE24090_SE04				Rev. 1		Data : 31/03/2025				Sheet : 1 of : 21		

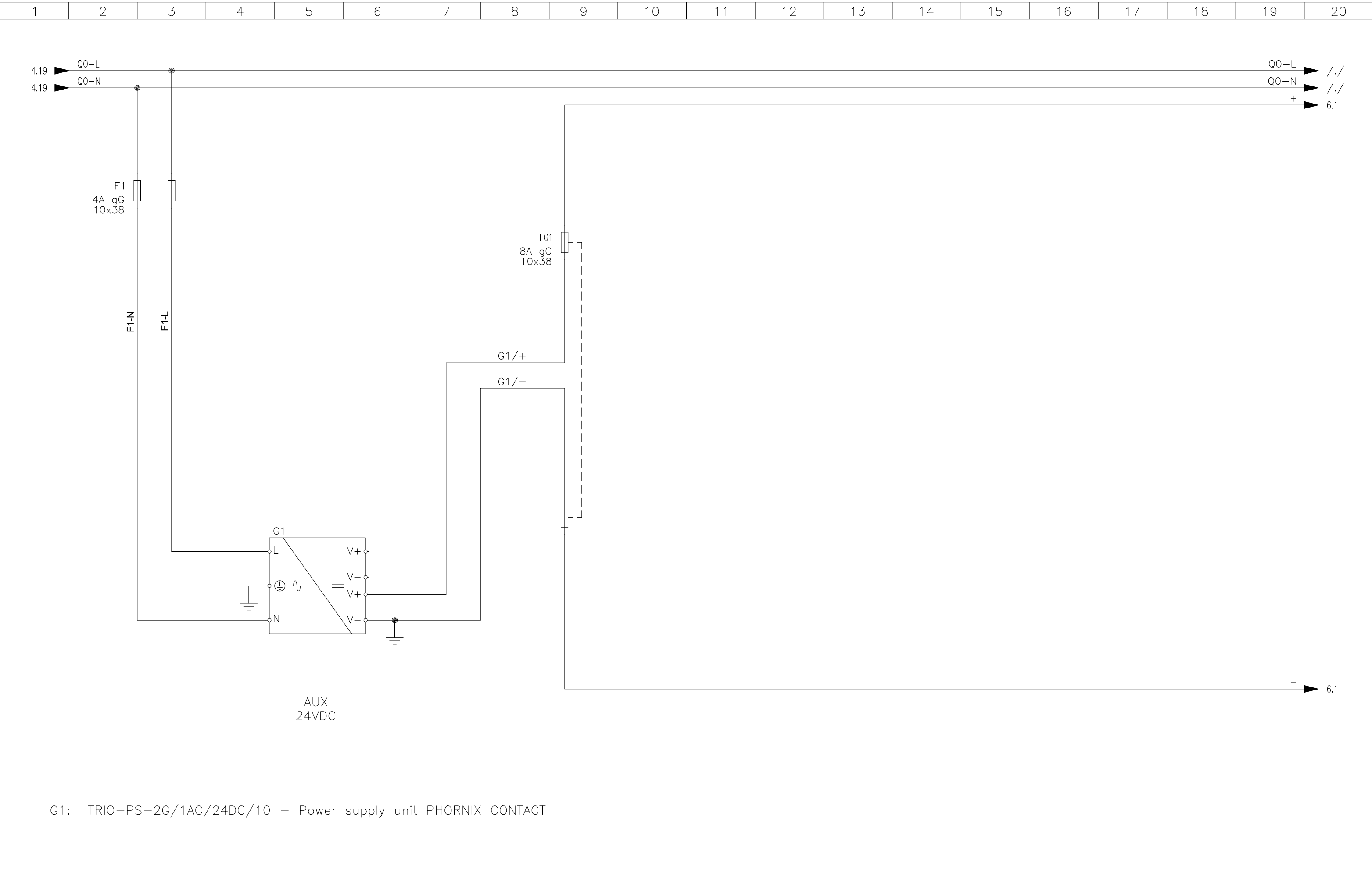
CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

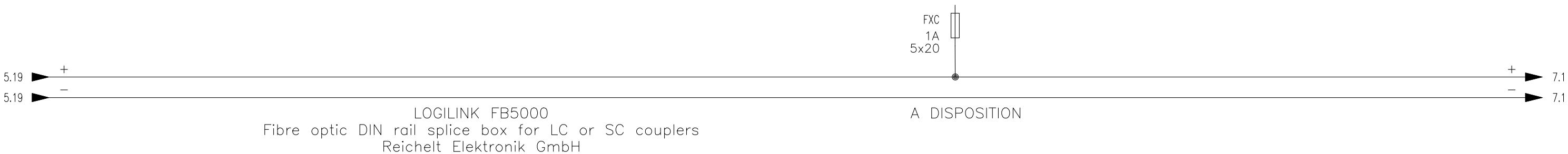
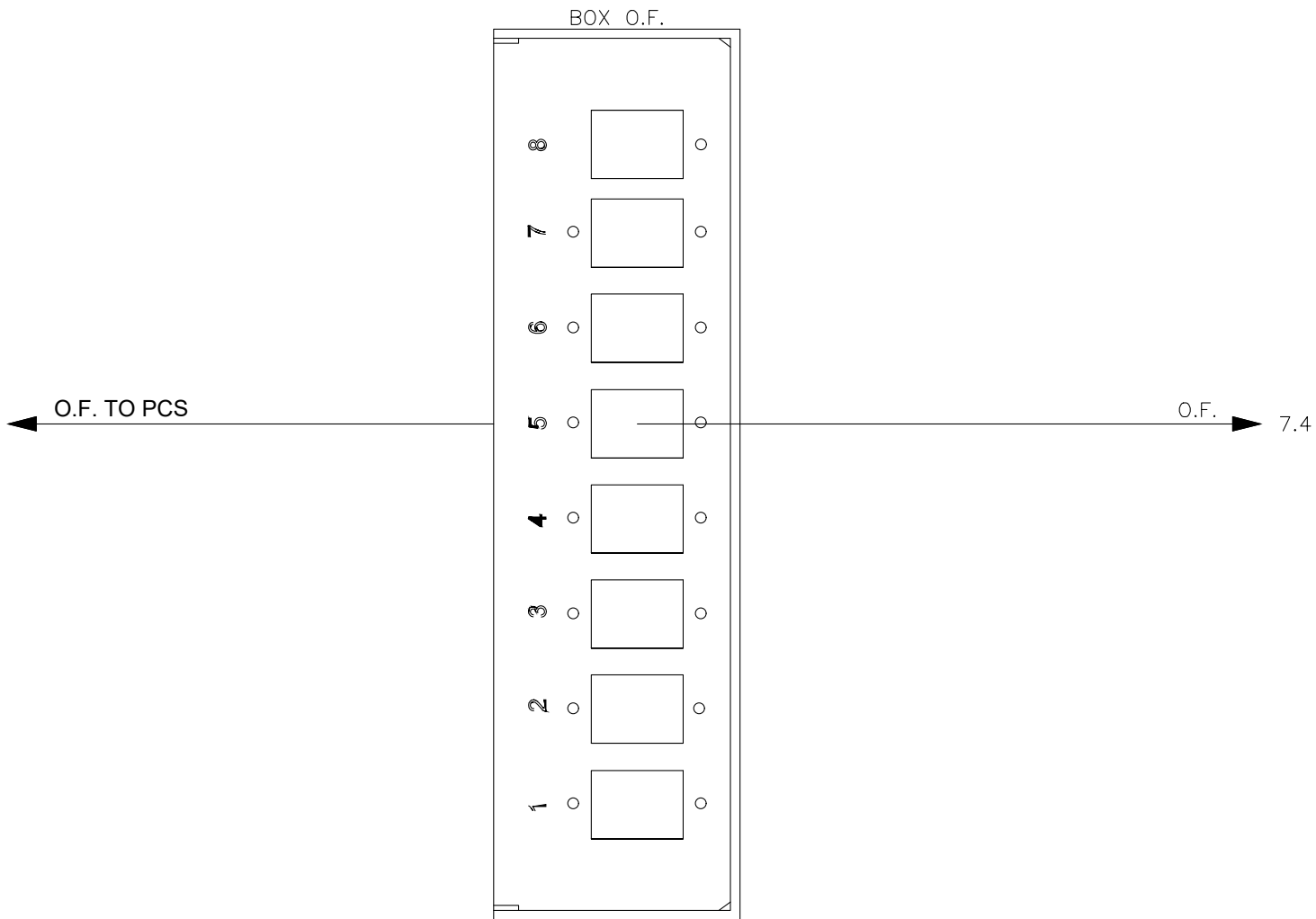
	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE



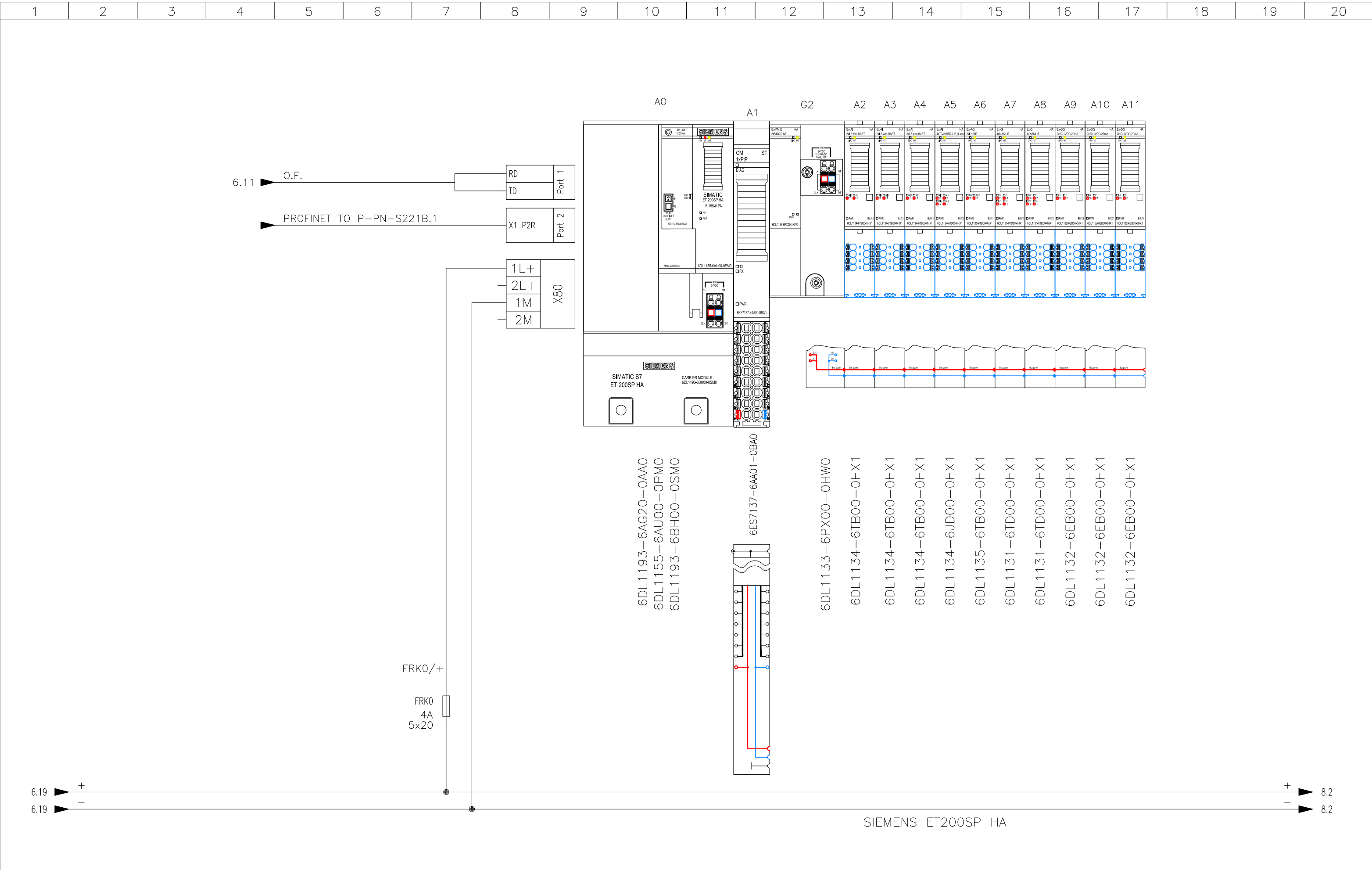




LOGILINK FB5000
Fibre optic DIN rail splice box for LC or SC couplers
Reichelt Elektronik GmbH

A DISPOSITION

General Contractor : Manufacture : 		Project : P-PN-S222B.1 LNG LOADING BAY S-222B	Customer :  Contract : IE24090 S/N : IE24090_SE04	Object : Rev. 2 Data : 11/04/2025 Rev. 1 Data : 31/03/2025		By :	
						Date : 26/03/2024	Sheet : 6 of : 21



7.19 +
7.19 -

9.1 +
9.1 -

F2
1A
5x20

T/RA- 9.12
T/RB+ 9.12

11 12 13 14 15 16

L+ M

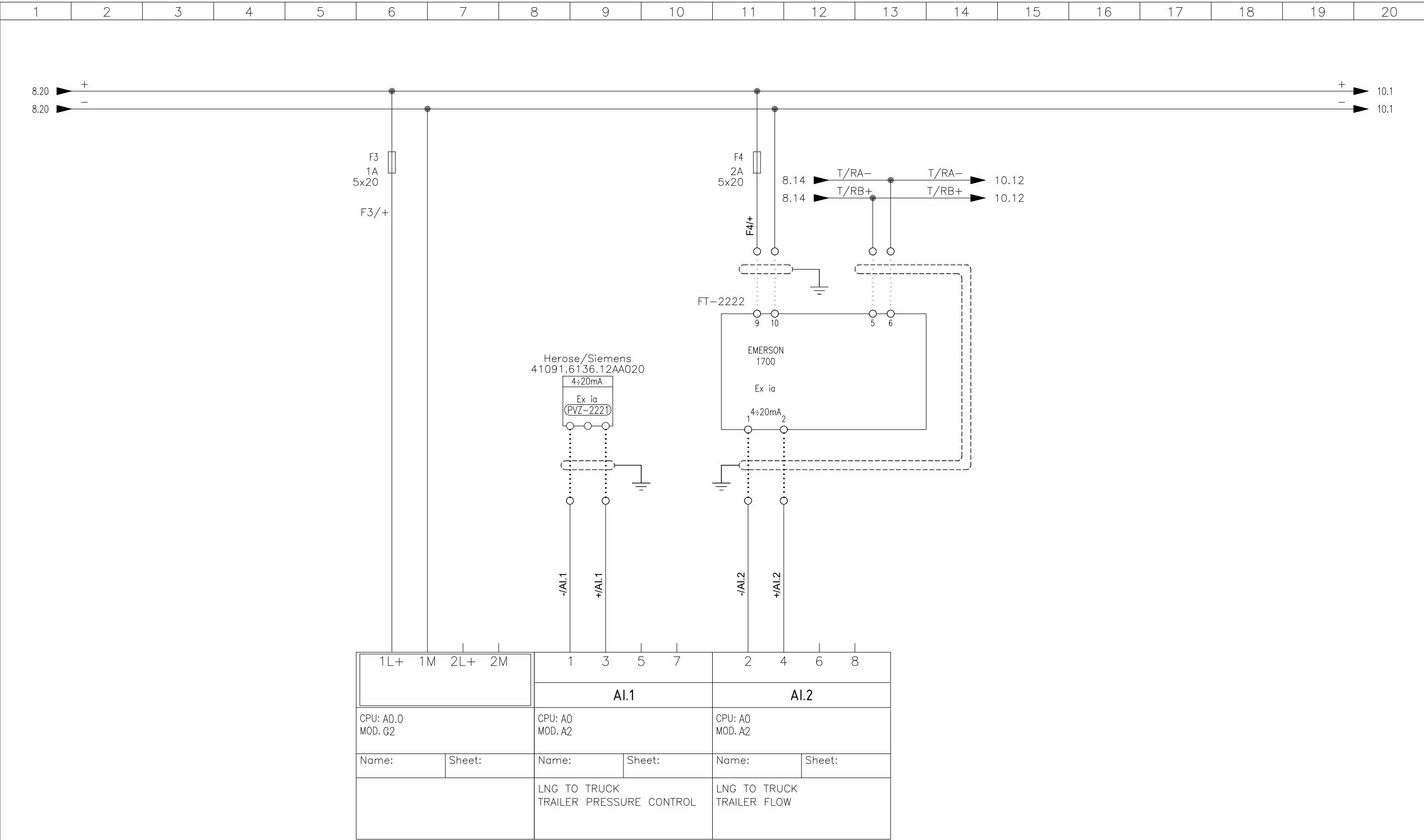
CPU: A0
MOD. A1

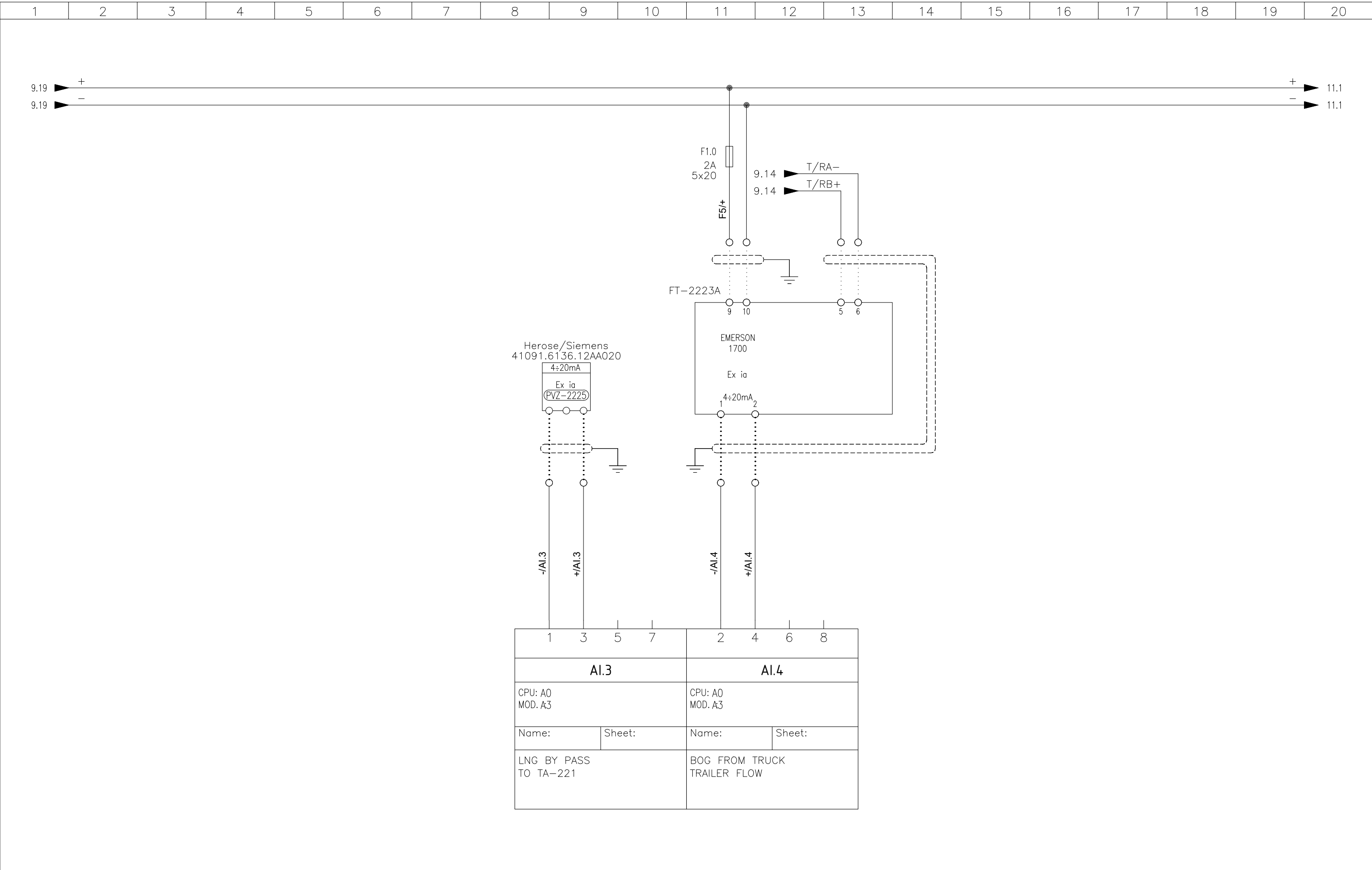
Name: Sheet:

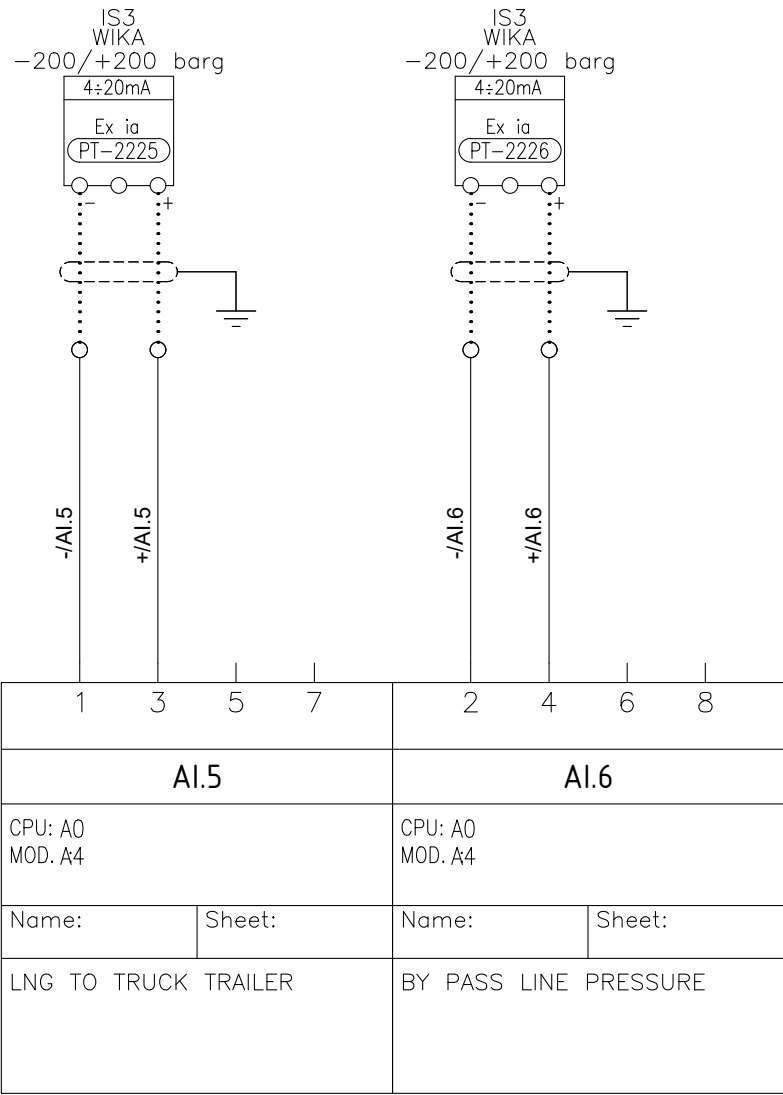
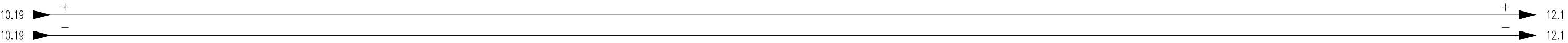
RS-485
TO FLOW TRANSITTER

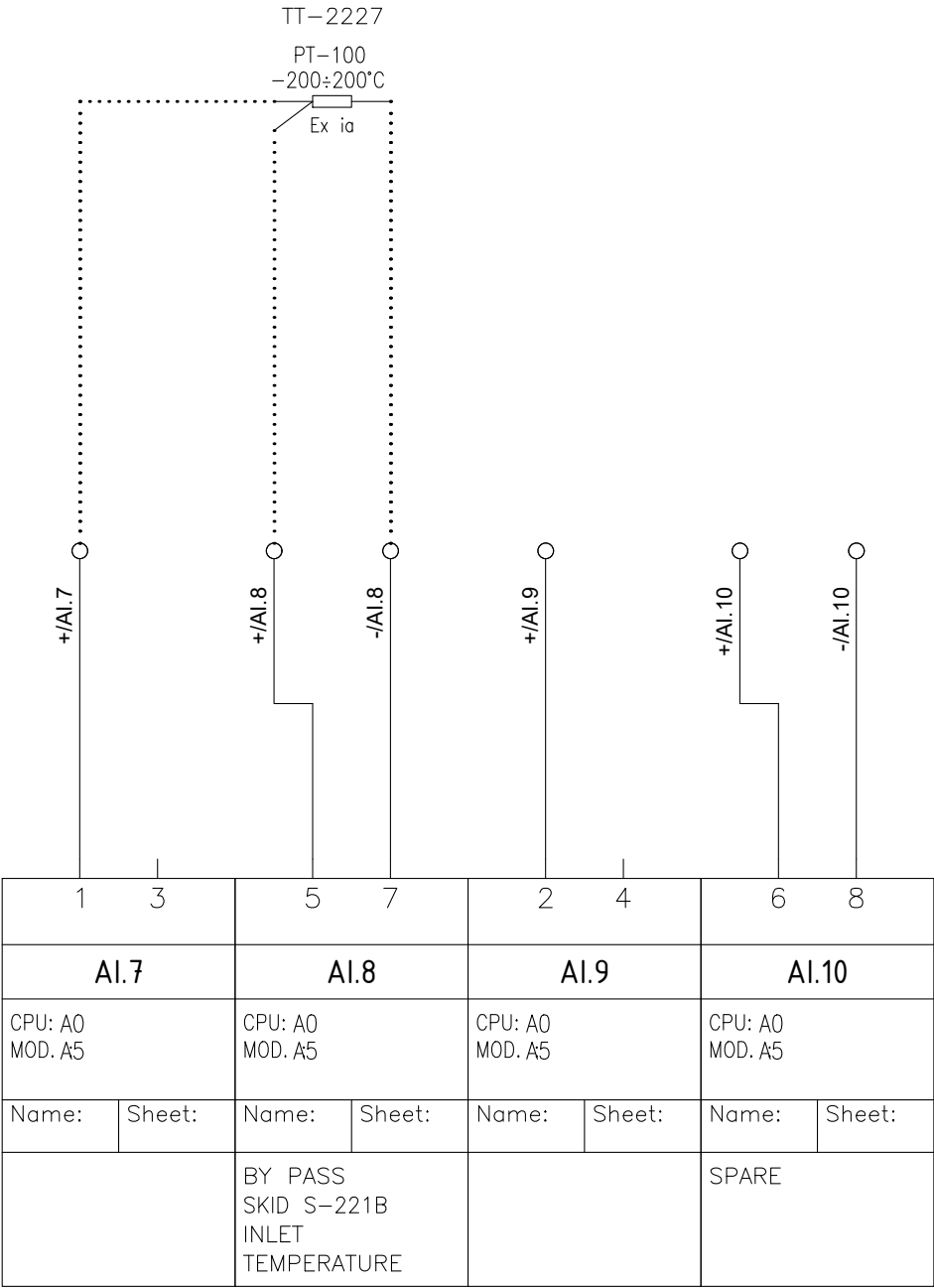
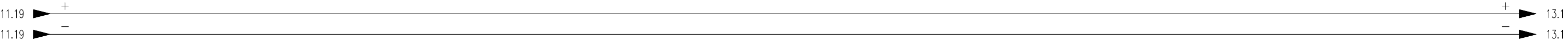
Name:					Sheet:				
CPU: A0 MOD. A1									
12 1 2 3 4 5 6 7 8 9/10									

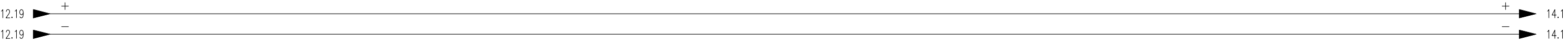
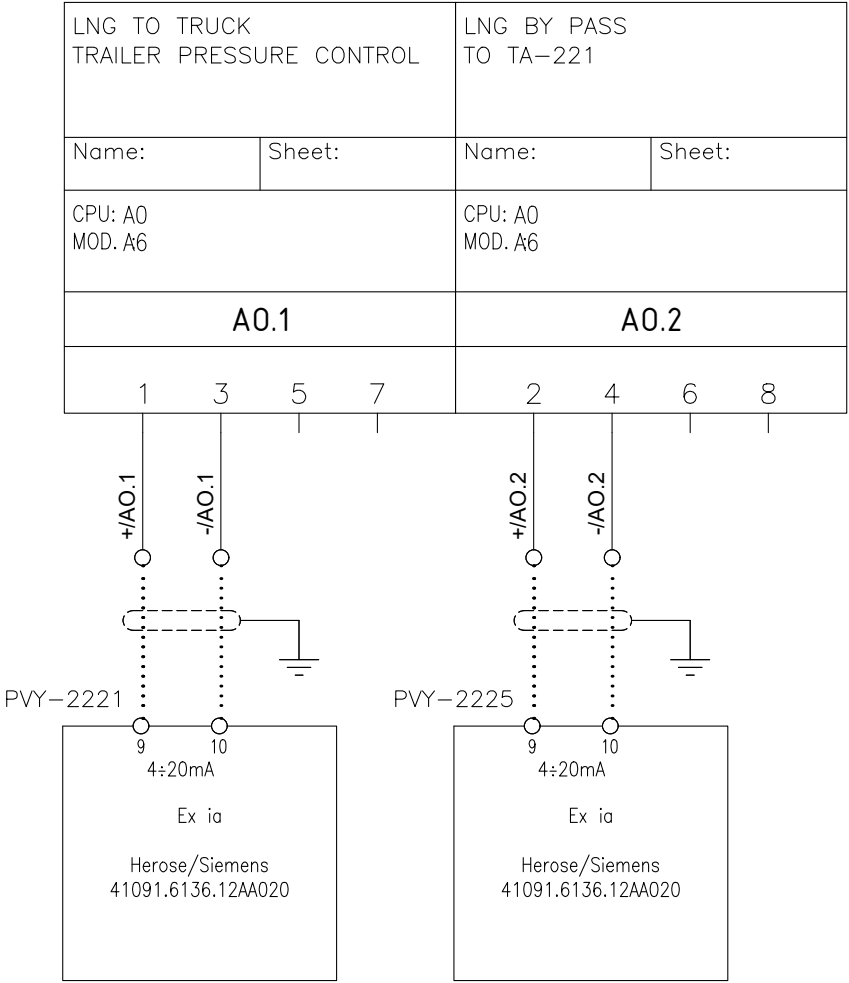
General Contractor :		Manufacture : 	Project : P-PN-S222B.1 LNG LOADING BAY S-222B	Customer : 	Object :			
					By :			
				Contract : IE24090	Rev. 2	Data : 11/04/2025	Date : 26/03/2024	
				S/N : IE24090_SE04	Rev. 1	Data : 31/03/2025	Sheet : 8	of : 21









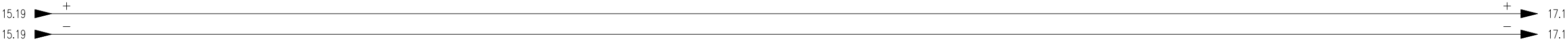


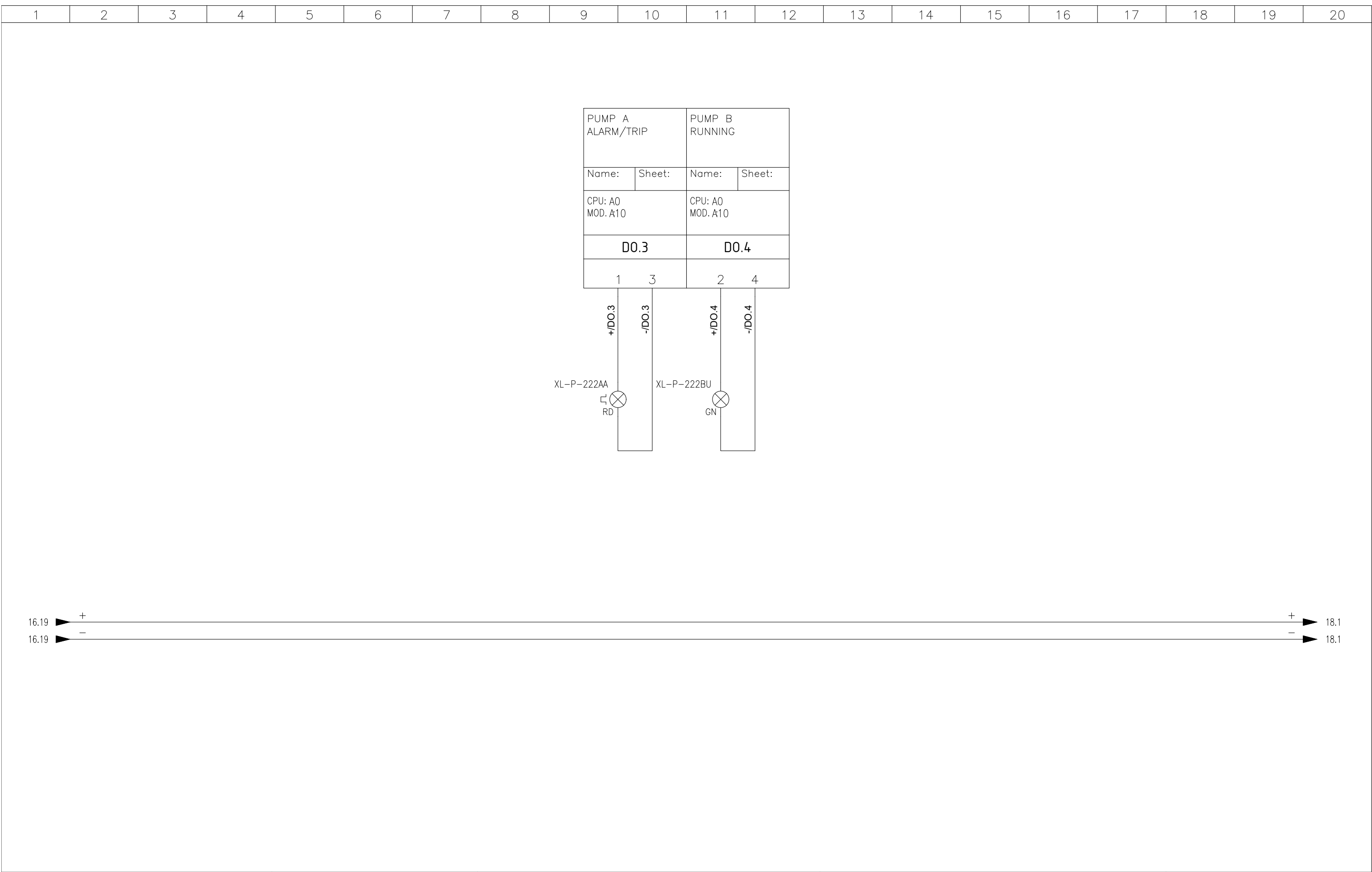
[illegible]

General Contractor :		Manufacture : 	Project : P-PN-S222B.1 LNG LOADING BAY S-222B	Customer : 	Object :			
					By :			
			Contract : IE24090		Rev. 2	Data : 11/04/2025	Date : 26/03/2024	
			S/N : IE24090_SE04		Rev. 1	Data : 31/03/2025	Sheet : 14	of : 21

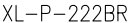
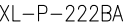
[illegible]

General Contractor : Manufacture : 		Project : P-PN-S222B.1 LNG LOADING BAY S-222B	Customer :  Contract : IE24090 S/N : IE24090_SE04	Object :		By :	
				Rev. 2 Data : 11/04/2025	Date : 26/03/2024	Rev. 1 Data : 31/03/2025	Sheet : 15 of : 21





PUMP B READY		PUMP B ALARM/TRIP	
Name:	Sheet:	Name:	Sheet:
CPU: A0 MOD. A11		CPU: A0 MOD. A11	
D0.5		D0.6	
1	3	2	4



Project :

P-PN-S222B.1

LONG LOADING BAY S-222B

Customer :



Contract : IE24090

IE24090_SE04

Object :

Rev. 2

Data : 11/04/2025

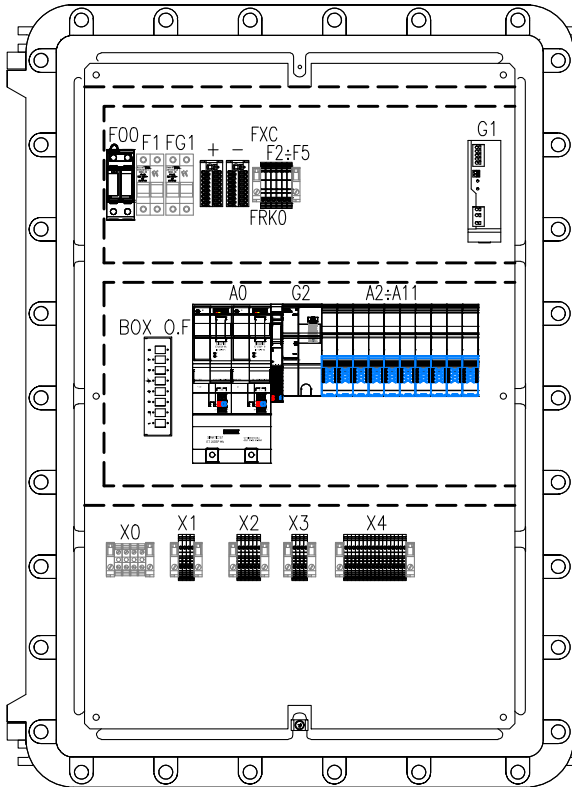
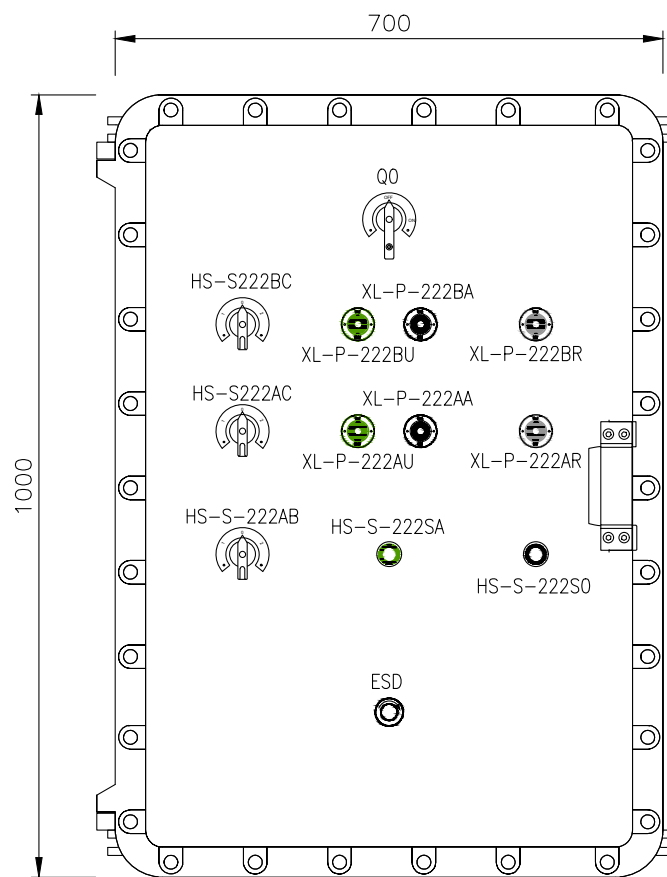
Rev. 1

Data : 31/03/2025

By :

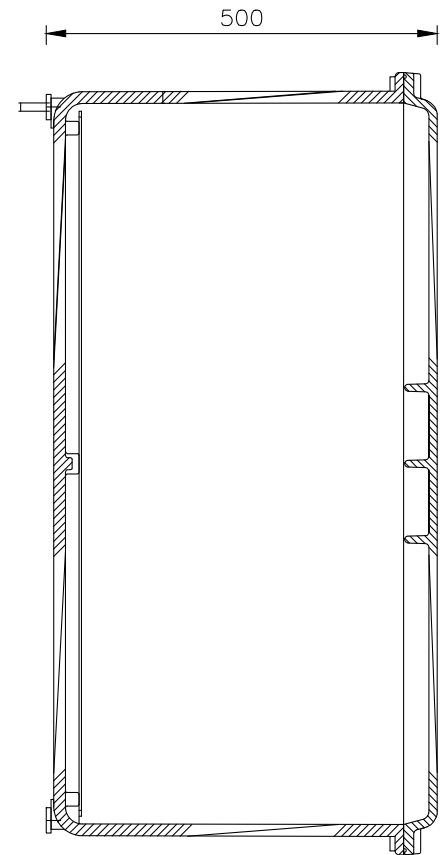
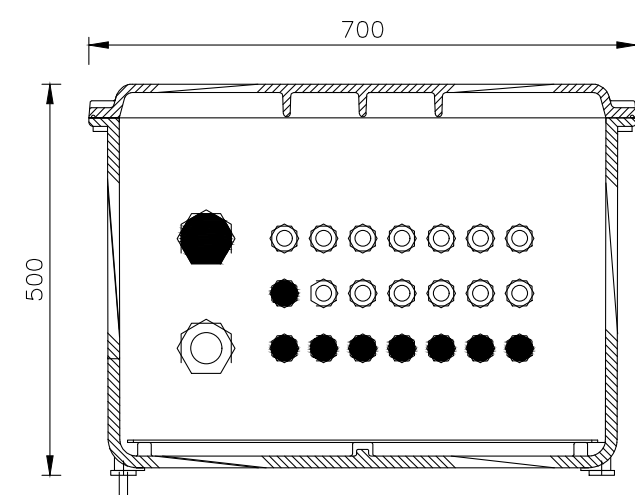
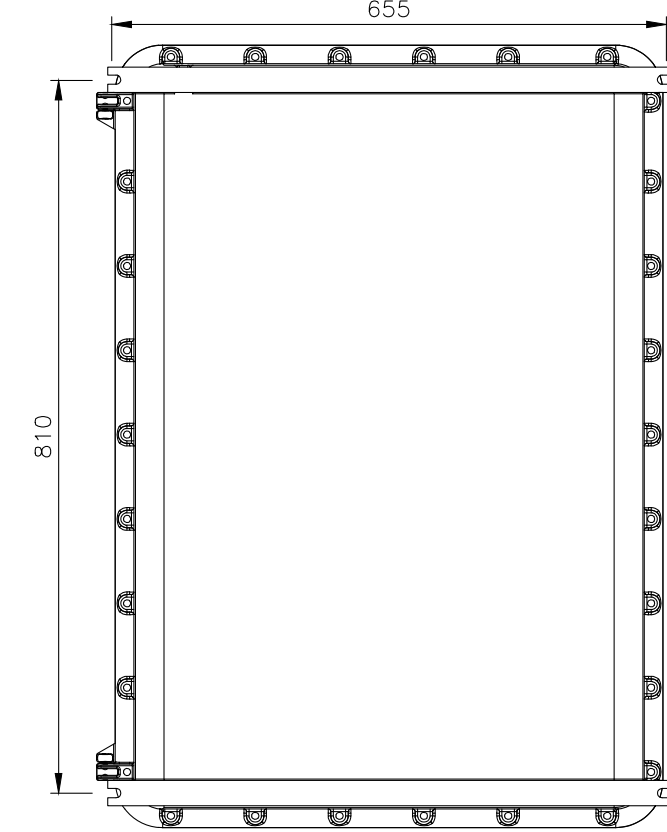
Date : 26/03/2024

Sheet : 18 of : 21



NOTE

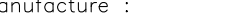
- Type : EJB-7
- Ex-d [ia] IIB T6 IP65 ATEX -20/+50°C
- Installation area is zone 2 IIA T1
- Weight 350kg
- Power dissipation 81W



1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----

INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
–	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
+	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
A0	6DL1193–6MD00–0AA0	"GUIDA PROFILATA 1500MM (59""")"	SIEMENS
A0	6DL1155–6AU00–OPM0	ET 200SP HA, IM155–6 PN	SIEMENS
A0	6DL1193–6AR00–0AA0	ET 200SP HA, ADATTATORE DI BUS BA 2XRJ45	SIEMENS
A0	6DL1193–6BH00–0SM0	MODULO SUPPORTO IM SINGLE	SIEMENS
A1	6ES7137–6AA01–0BA0	ET 200SP, CM PTP, VPE 1	SIEMENS
A1	6ES7193–6BP00–0DA0	BaseUnit BU15–P16+A0+2D, BU type A0, push–in terminals,	SIEMENS
A10	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A10	6DL1132–6EB00–0HX1	ET 200SP HA, EX–DQ 2x23,1VDC/20MA	SIEMENS
A11	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A11	6DL1132–6EB00–0HX1	ET 200SP HA, EX–DQ 2x23,1VDC/20MA	SIEMENS
A2	6DL1193–6BP00–0DW0	ET 200SP HA, Ex–BU TYPE W0	SIEMENS
A2	6DL1134–6TB00–0HX1	ET 200SP HA, EX–AI 2xI 2–WIRE HART	SIEMENS
A3	6DL1193–6BP00–0DW0	ET 200SP HA, Ex–BU TYPE W0	SIEMENS
A3	6DL1134–6TB00–0HX1	ET 200SP HA, EX–AI 2xI 2–WIRE HART	SIEMENS
A4	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A4	6DL1134–6JD00–0HX1	ET 200SP HA, EX–AI 4xTC/2xRTD 2–/3–/4–W	SIEMENS
A5	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A5	6DL1134–6JD00–0HX1	ET 200SP HA, EX–AI 4xTC/2xRTD 2–/3–/4–W	SIEMENS
A6	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A6	6DL1135–6TB00–0HX1	ET 200SP HA, EX–AQ 2xI HART	SIEMENS
A7	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A7	6DL1131–6TD00–0HX1	ET 200SP HA, EX–DI 4xNAMUR	SIEMENS
A8	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A8	6DL1131–6TD00–0HX1	ET 200SP HA, EX–DI 4xNAMUR	SIEMENS
A9	6DL1193–6BP00–0BX1	ET 200SP HA, Ex–BU TYPE X1	SIEMENS
A9	6DL1132–6EB00–0HX1	ET 200SP HA, EX–DQ 2x23,1VDC/20MA	SIEMENS
BOX O.F.	LOGILINK FB5000	Fibre optic DIN rail splice box for LC or SC couplers	Reichelt Elektronik GmbH
ESD	M–0445	RED EMERGENCY PUSH–BUTTON , TURN TO RELEASE, 1 N.O. + 1 N.C. CONTACTS	CORTEM
F00	952110	DG M TT 2P 275 – SPD	DEHN
F1	2302038	Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V	ITALWEBER
F1	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F4	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F5	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FG1	2301138	Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V	ITALWEBER
FRK0	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FXC	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
G1	2903149	TRIO–PS–2G/1AC/24DC/10 – Power supply unit	PHOENIX CONTACT
G2	6DL1133–6PX00–0HW0	ET 200SP HA, Ex–PM E POWER MODULE	SIEMENS
HS–S–222AB	M–0553	EXTERNAL HANDLE	CORTEM
HS–S–222AB	U2–10E/R	"SONTHEIMER SEL.SWITCH ""1–0–2""",2P,10A, WITH 3 FIXED POSITIONS"	CORTEM
HS–S222AC	M–0553	EXTERNAL HANDLE	CORTEM
HS–S222AC	WS2–10E/R	"SONTHEIMER SEL.SWITCH ""1–2""",2P,10A, WITH 2 FIXED POSITIONS (SEE DETAIL)"	CORTEM
HS–S222BC	M–0553	EXTERNAL HANDLE	CORTEM
HS–S222BC	WS2–10E/R	"SONTHEIMER SEL.SWITCH ""1–2""",2P,10A, WITH 2 FIXED POSITIONS (SEE DETAIL)"	CORTEM
HS–S–222S0	M–0429/R	RED PUSH–BUTTON 1 N.O. + 1 N.C.	CORTEM
HS–S–222SA	M–0429/V	GREEN PUSH–BUTTON 1 N.O. + 1 N.C.	CORTEM
P–PN–S222B.1	EJB–7	ENCLOSURE 1000x700x510 mm L x H x D	CORTEM
P–PN–S222B.1	BFE–7	INTERNAL FRAME	CORTEM
Q0	M–0553L	EXTERNAL HANDLE, PADLOCKABLE	CORTEM
Q0	A3–10E/R	"SONTHEIMER NOT AUTOMATIC SWITCH ""ON–OFF""",3P,10A, PADLOCKABLE IN ""OFF"" POSITION"	CORTEM
XL–P–222AA	M–0457/3R24	SIGNALLING LED RED COLOUR 24VDC	CORTEM
XL–P–222AR	M–0457/3I24	SIGNALLING LED WHITE COLOUR 230V	CORTEM
XL–P–222AU	M–0457/3V24	SIGNALLING LED GREEN COLOUR 24VDC	CORTEM
XL–P–222BA	M–0457/3R24	SIGNALLING LED RED COLOUR 24VDC	CORTEM
XL–P–222BR	M–0457/3I24	SIGNALLING LED WHITE COLOUR 230V	CORTEM

INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
XL-P-222BU	M-0457/3V24	SIGNALLING LED GREEN COLOUR 24VDC	CORTEM

General Contractor : Manufacture : 		Project : P-PN-S222B.1 LNG LOADING BAY S-222B	Customer : 	Object :		By :	
				Rev. 2	Data : 11/04/2025	Date : 26/03/2024	
		Contract : IE24090	Rev. 1	Data : 31/03/2025	Sheet : 20	of : 21	
		S/N : IE24090_SE04					

